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SUPERSEDING
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April 2, 1975

FEDERAL SPECIFICATION

CLEANING COMPOUND, ALKALI, PIPELINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers the requirements for an alkali type dry granular compound, suitable for cleaning drain lines (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

O-S-598 - Sodium Hydroxide, Technical.
O-S-634 - Sodium Nitrate, Technical (Nitrate of Soda).
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substance Act.
21 CFR 295 - Poison Prevention Packaging Act of 1970.
49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The compound shall be a uniform, free-flowing, granular type compound. It shall contain only those materials listed in table I.

TABLE I. Materials

Ingredients	Percent by Weight
Flake sodium hydroxide	54 ± 5
Sodium nitrate	30 ± 3
Sodium chloride	11 ± 1
Aluminum cuttings	4.2 ± 0.4

3.1.1 Sodium hydroxide. The flake sodium hydroxide shall comply with the chemical requirements of O-S-598, when tested as specified in 4.3.3, and shall meet the following sieve requirements when tested as specified in 4.3.2:

U.S. Sieve No.	Percent Through
4 (4.75 mm)	100
20 (850 µm)	20 max.
40 (425 µm)	10 max.

3.1.2 Sodium nitrate. The sodium nitrate shall comply with the chemical and physical requirements of O-S-634 when tested as specified in 4.3.4.

3.1.3 Sodium chloride. The sodium chloride shall contain not less than 97.5 percent sodium chloride and shall have not more than 0.5 percent moisture. The sodium chloride shall meet the following sieve requirements when tested as specified in 4.3.2:

U.S. Sieve No.		Requirement
Through	Retained On	percent
	8 (2.36 mm)	Trace
8 (2.36 mm)	12 (1.70 mm)	2.5 (max)
12 (1.70 mm)	16 (1.18 mm)	20.0 (max)
50 (300 µm)	70 (212 µm)	15.0 (max)
70 (212 µm)		15.0 (max)

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3.2 Sieve size. The finished compound shall pass a U.S. No. 3-1/2 (5.6 mm) sieve completely when tested as specified in 4.3.2.

3.3 Heat of reaction. The compound, when tested as specified in 4.3.1, shall produce a temperature rise of not less than 78 degrees C.

3.4 Solubility. The compound, when tested as specified in 4.3.1, shall have no insoluble alkali cake or aluminum cuttings at the end of 10 minutes. A black flocculent precipitate (impurities in aluminum) shall not be considered as failing this test.

3.5 Foam head. The foam head developed during the reaction specified in 4.3.1 shall not exceed 2 inches (5 cm).

3.6 Labeling. The unit container label shall be in accordance with the Federal Hazardous Substances Act (see 5.3).

3.7 Workmanship. The compound shall contain no foreign material, lumpy material, or aluminum chain. The compound shall be granular type and free of objectionable odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Sampling and inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated herein.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all the containers of the compound offered for delivery at one time.

4.2.2 Examination of filled containers. From each lot, containers shall be selected at random in accordance with MIL-STD-105 at inspection level II, and shall be examined for conformance with the applicable requirements of this specification. Any containers having one or more defects or under required fill shall be rejected. The acceptable quality level (AQL) shall be 1.5 percent defective. Defects shall be classified as specified in table II.

TABLE II. Examination of filled containers

Examine	Defect
Container	Not specified type or size container. Any evidence of leakage or corrosion from any part of the container. Any hole, tear, puncture, scuff, dent, or other defect in body of container. Closures not as specified.
Labeling	Printing or stenciling pertaining to the applicable information and instructions for end item use of the specified material omitted, illegible, incorrect, incomplete, smeared.
Weight	Net weight not as specified.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored in accordance with table III. The sample unit shall be one container fully prepared for delivery, selected just prior to the closing operation. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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TABLE III. Examination of preparation for delivery

Examine	Defect
Container	Not as specified.
Contents	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
	Cushioning inadequate or improper.

4.2.3.1 Examination of closure, waterproofing, and banding of shipping containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix of that specification.

4.2.4 Acceptance tests.

4.2.4.1 Chemical and physical analyses. Sample containers shall be selected in accordance with MIL-STD-105 at inspection level S-2, and tested in accordance with 4.3. If any container specimen fail in any test, the lot represented shall be rejected.

4.3 Test methods.

4.3.1 Heat of reaction test.

4.3.1.1 Equipment. A 30 x 3.8 cm pyrex test tube shall be fitted with a one-hole slotted cork stopper. A thermometer shall be inserted through the stopper and extended into the test tube in such a manner that when 100 ml of water is in the test tube, the bulb shall be approximately at the 50 ml line. The test tube shall be firmly clamped in an upright position.

4.3.1.2 Procedure. Add 100 ml of distilled water at a temperature of 15 degrees C to 20 degrees C to the test tube. Record the temperature of the water. Quickly weigh out 30 gr. ± 0.1 gr. of the compound and pour into the 100 ml of water. Reinsert cork and thermometer and record the maximum temperature rise (see 3.3). (Highest temperature recorded minus the initial temperature.) During the reaction, the maximum foam head shall not exceed 2 inches (5 cm) (see 3.5), and there shall be active churning of the aluminum particles. At the end of 10 minutes, the test tube and contents shall be examined for compliance with 3.4.

4.3.2 Sieve tests. All sieve tests shall be run in accordance with the applicable test procedures outlined in Fed. Test Method Std. No. 536.

4.3.3 Sodium hydroxide. The sodium hydroxide chemical analysis shall be in accordance with methods as specified in O-S-598.

4.3.4 Sodium nitrate. The sodium nitrate chemical and physical analyses shall be in accordance with methods as specified in O-S-634.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Twelve (12) ounces of the compound shall be packaged in a container specified in 5.1.1.1 or 5.1.1.2 and in accordance with the Poison Prevention Packaging Act. The container closure shall have a reclosable feature.

5.1.1.1 Metal can. The metal can shall be round with both ends crimped or double-seamed in position. The can shall be made throughout from tin plate, terneplate, or blackplate, or any combination of these metals as recommended by the manufacturer for the product being packaged.

5.1.1.2 Plastic container. The plastic container shall be of the type and size commonly used by the industry.

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5.1.2 Level C. The compound shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve cans of the compound, packaged as specified in 5.1.1, shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve cans of the compound shall be packed as specified in 5.2.1 except the box shall be class domestic.

5.2.3 Level C. The compound, packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification rules, as applicable.

5.3 Labeling and marking. In addition to any special marking required in the procurement documents, the unit and shipping container shall be labeled and marked in accordance with 3.6, 5.3.1, and Fed. Std. No. 123 for civil agencies, and MIL-STD-129 for military agencies, as applicable (see 6.2).

5.3.1 Precautionary marking. Exterior shipping containers shall be marked in accordance with Department of Transportation requirements. Labels for 12-ounce containers shall conform to the Federal Hazardous Substance Act (FHS). Labels shall be in bold type and securely affixed to the container and shall read as follows:

POISON

Precautions - Read Before Using

DANGER Contains sodium hydroxide (caustic soda lye). Corrosive: causes severe eye and skin damage. Keep material and solutions of material away from eyes, skin, and clothing; don't get it on you. Always keep out of reach of children. Harmful if swallowed.

ANTIDOTES Eyes - immediately flood with water for at least 20 minutes. Follow with 5 percent Boric Acid solution. INTERNAL - if swallowed, do not induce vomiting. Give vinegar or juice of lemon, orange, or grapefruit freely. Follow with olive oil, butter, or other cooking oil. EXTERNAL - flood with water, then wash with vinegar.

CALL PHYSICIAN AT ONCE

6. NOTES

6.1 Intended use. Cleaning compound, alkali, pipeline is intended for opening clogged drain lines and as a regular maintenance compound for keeping drain lines sanitary and trouble free. It is not intended for use with garbage disposal or dishwasher.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Applicable levels of packaging and packing required (see 5.1 and 5.2).
(c) Applicable marking required (see 5.3).

6.3 Basis of purchase. The compound should be purchased by net weight.

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MILITARY INTEREST:

Custodian:

Air Force - 68

CIVIL AGENCY COORDINATING ACTIVITIES:

CPSC
GSA-FSS
HEW-FDA
VA-DMS

PREPARING ACTIVITY: GSA-FSS

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